

6 Organisms can be classified into one of three domains.

(a) Organisms belonging to two of these domains have prokaryotic cells.

(i) Bacteria are one of these domains.

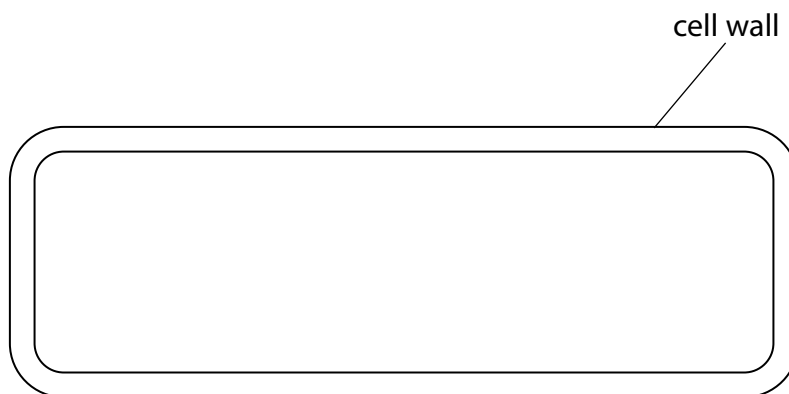
Name the other domain that has prokaryotic cells.

(1)

(ii) The diagram shows the outline of a bacterial cell.

Draw **three** labelled features on this diagram that may be found in a prokaryotic cell.

(3)



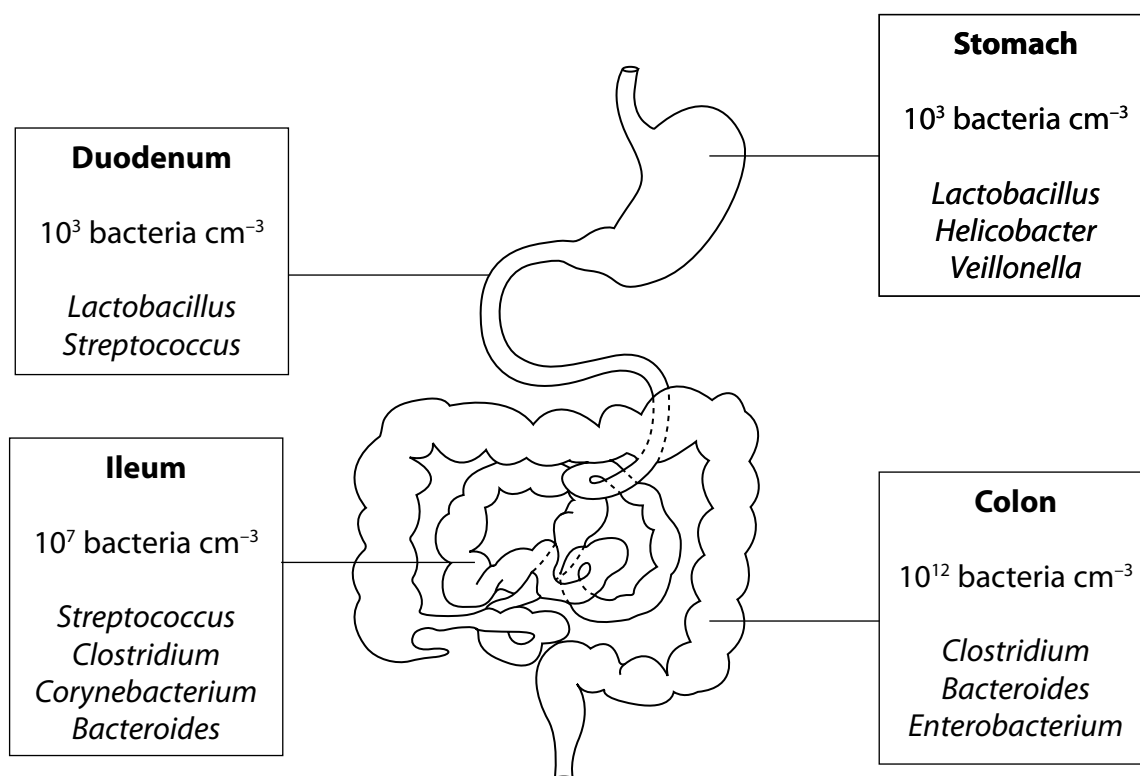
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*(b) A variety of different types of bacteria is found in the human digestive system.

The diagram shows part of the human digestive system and the number and types of bacteria that can be found in each organ.



The table gives some information about conditions in the digestive system.

Organ	pH	Oxygen content
Stomach	1 to 3	High ↓ Low
Duodenum	6 to 7	
Ileum	6 to 8	
Colon	5 to 7	

Explain the distribution of bacteria in the digestive system. Use the information in the diagram and table to support your answer.

(6)

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(Total for Question 6 = 10 marks)